

## SUMMARY TABLE: PHYSICS GRADE 10

### Sub-Strand 4.2: Introduction to Space Physics — Teacher Reference (Pre-filled)

| SUMMARY TABLE: PHYSICS GRADE 10 |                                                                                                                                                                                           |
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| Sub-Strand                      | Sub-Strand 4.2: Introduction to Space Physics                                                                                                                                             |
| Driving Question                | DRIVING QUESTION: How did everything we see in the night sky above Nairobi come to exist, and how do scientists gather evidence about a universe they cannot directly touch or travel to? |

### INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

| Lesson # and Title                                                                                 | What did I observe?                                                                                                                                                                                                                                                                                                                                                                                                           | What did I learn?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | How does this explain the phenomenon?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>Lesson 1</b><br><b>The Big Bang Theory: Launching the Phenomenon</b>                            | A high-resolution image of the night sky above the Rift Valley showing the Milky Way, Moon, and visible planets; a Kurzgesagt animation showing the Big Bang and early universe expansion; a Khan Academy article describing the Big Bang, the expansion of the universe, and the timeline from singularity to the present day.                                                                                               | The universe began approximately 13.8 billion years ago from an infinitely hot, infinitely dense singularity in an event called the Big Bang. The universe has been expanding and cooling ever since. Everything observable in tonight's sky — stars, planets, galaxies, even space and time itself — originated from this event. Scientists use evidence such as the expansion of galaxies and the Cosmic Microwave Background Radiation (CMB) to support this theory, even though no human was present and we cannot travel back in time to observe it directly. | The night sky above Nairobi appears permanent and unchanging, but this is because the time scales of cosmic change are far longer than a human lifetime or even recorded human history. The Big Bang Theory explains the origin of everything observable in the universe — not as an explosion within existing space, but as the expansion of space itself from a singularity. The tension between the static appearance of the night sky and the scientific evidence of a dynamic, expanding universe is the central puzzle of this unit, and every subsequent lesson will add evidence to help explain and model this phenomenon. |
| <b>Lesson 2</b><br><b>Evolution of the Universe: From the Big Bang to the Structured Night Sky</b> | Learners observed: (1) a physical rope timeline showing 12 key cosmic events across 13.8 billion years, making the immense scale of cosmic time tangible; (2) a gallery of real galaxy images (elliptical, spiral, barred spiral, irregular) classified using NASA's galaxy classification framework; (3) a short video ('Evolution of the Universe in 5 Minutes') visualising the sequential stages from the Big Bang to the | Learners learned that: (1) the universe began approximately 13.8 billion years ago in the Big Bang — before which neither matter, energy, space, nor time existed as we know them; (2) the universe evolved through distinct stages: quark-gluon plasma → protons and neutrons → first atoms (hydrogen and helium, ~380,000 years after Big Bang) → first stars (~200 million years) → first galaxies (~1 billion years) → our                                                                                                                                     | Learners explained that: (1) the apparently static and eternal night sky above Nairobi is in fact the product of 13.8 billion years of dynamic cosmic evolution — from a hot, dense, structureless state to the complex, galaxy-filled universe visible tonight; (2) the structured universe (stars, galaxies, planets) emerged progressively as matter cooled and gravity caused it to clump together over vast timescales; (3) scientists                                                                                                                                                                                         |

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|                                                                                            | formation of galaxies, stars, and planetary systems.                                                                                                                                                                                                                                                                                                                           | Milky Way (~13.6 billion years ago) → our Solar System (~4.6 billion years ago); (3) the Milky Way — the glowing band visible from the Rift Valley and Lake Victoria — is a barred spiral galaxy containing over 200 billion stars, including our Sun; (4) the Cosmic Microwave Background radiation is one of the key pieces of evidence scientists use to confirm the Big Bang model; (5) the universe continues to expand today.                                                                                 | gather evidence for this evolutionary history through CMB radiation, galaxy observation and classification, and the measured expansion of the universe — without needing to physically travel to or touch the objects they study; (4) each star visible from Nairobi is itself a product of this evolutionary process, and the light reaching our eyes left those stars millions or billions of years ago.                                                                                                                                                                                                        |
| <b>Lesson 3</b><br><b>Classification of Celestial Bodies: Reading the Kenyan Night Sky</b> | Using Stellarium set to Nairobi's coordinates, learners identified and logged at least eight distinct celestial objects (including the Moon, Venus, Jupiter, Sirius, and the Milky Way band) by consulting each object's Stellarium data panel for type, distance, and magnitude evidence.                                                                                     | Learners constructed a six-class Classification Table (stars, planets, moons, comets, asteroids, galaxies) linking each class to its defining characteristics, its method of identification in Stellarium, and its approximate first-appearance era in the 13.8-billion-year cosmic timeline established in Lesson 2.                                                                                                                                                                                               | Learners revised their cumulative night-sky models with colour-coded class labels and cosmic-era annotations, and wrote a group Big Claim statement explaining that the variety of objects visible in the Nairobi night sky exists because different classes of objects formed at different stages of the universe's evolutionary history — advancing the class's collective answer to the driving question.                                                                                                                                                                                                      |
| <b>Lesson 4</b><br><b>The Solar System: Structure and Scale</b>                            | I walked the scaled Solar System in the corridor and observed that all four inner planets fit within the first 3 squares of toilet paper while Neptune required approximately 60 squares, revealing that the Solar System is overwhelmingly empty space and that the outer planets are separated by enormous distances.                                                        | I learned that the Solar System has eight planets arranged in two distinct zones separated by the asteroid belt, that distances are measured in astronomical units, and that gravitational force from the Sun is the organising principle that keeps all planets in orbital motion rather than flying off into space.                                                                                                                                                                                               | This explains the phenomenon because the planets visible above Nairobi such as Venus and Jupiter are part of a structured, gravity-governed system that formed from the same solar nebula described in Lesson 2, connecting what we see in the night sky to the cosmic history established by the Big Bang and stellar evolution.                                                                                                                                                                                                                                                                                 |
| <b>Lesson 5</b><br><b>Planetary Motion and Kepler's Laws</b>                               | Using the PhET 'Gravity and Orbits' simulation, learners observed that planetary orbits are ellipses (not perfect circles), that planets move faster when closer to the Sun and slower when farther away, and that planets farther from the Sun take longer to complete one orbit. They also observed real orbital data for six Solar System planets (Mercury through Saturn). | Learners learned Kepler's Three Laws of Planetary Motion: (1) First Law — planets orbit the Sun in ellipses with the Sun at one focus; (2) Second Law — a planet sweeps out equal areas in equal times, meaning it speeds up near the Sun; (3) Third Law — $T^2 \propto r^3$ , meaning the square of a planet's orbital period is proportional to the cube of its semi-major axis. They mathematically verified the Third Law by calculating $T^2/r^3 \approx 1$ for Earth, Mars, Jupiter, and three other planets. | Using Kepler's Three Laws, learners can explain why Venus and Jupiter appear to move at different rates across the night sky above Nairobi — they are at different distances from the Sun and therefore have different orbital periods governed by $T^2 \propto r^3$ . The laws explain that the Solar System is a mathematically ordered system, not a random arrangement, and that its structure has been stable and predictable since its formation 4.6 billion years ago. The constant $T^2/r^3$ ratio is evidence that a single underlying force — yet to be fully explained — governs all planetary orbits. |
| <b>Lesson 6</b><br><b>Gravitational Forces in</b>                                          | Tidal height data from Mombasa Old Port showing two high tides and two low tides                                                                                                                                                                                                                                                                                               | Newton's Law of Universal Gravitation states that gravitational force is proportional                                                                                                                                                                                                                                                                                                                                                                                                                               | The night sky above Nairobi is not static or separate from life on Earth — the same                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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| <b>Space</b>                                                                                                                       | per day, and a diagram of Earth-Moon tidal bulges with Lake Victoria and Mombasa marked on the African continent.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | to the product of two masses and inversely proportional to the square of the distance between them. This universal force keeps the Moon orbiting Earth, Earth orbiting the Sun, and causes the Moon to pull Earth's water into tidal bulges that produce measurable twice-daily tides at Mombasa and detectable micro-tides on Lake Victoria.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | gravitational force that organises the Solar System reaches down to pull water at the Kenyan coast and on Lake Victoria, meaning that the universe visible above Nairobi is actively shaping everyday Kenyan environments right now.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Lesson 7</b><br><b>Telescopes: Windows to the Universe</b>                                                                      | In the night sky above Nairobi, thousands of stars, the Milky Way band, the Moon, and visible planets appear static and permanent. Yet every piece of scientific evidence we have discussed across seven lessons — the Big Bang, stellar evolution, galaxy formation, planetary orbits, gravitational effects — was gathered by scientists using telescopes and other remote-sensing instruments, not by travelling to these objects. Telescopes reveal that the apparently still night sky is in fact a window into billions of years of cosmic history, with each point of light representing a different moment in time depending on how far away it is. | Telescopes gather electromagnetic radiation — visible light, infrared, ultraviolet, or radio waves — that travels from distant objects to Earth, allowing scientists to study the universe without physically travelling to it. Refracting telescopes use convex lenses to bend and focus light; reflecting telescopes use concave mirrors to collect and focus light and can be built much larger. Radio telescopes use large dish antennas to detect radio waves emitted by galaxies, hydrogen clouds, and other astrophysical sources. Because light takes time to travel, telescopes looking at distant objects are also looking back in time — the JWST can observe light from galaxies that formed just 300 million years after the Big Bang. Africa participates in world-leading telescope science through the MeerKAT and SKA project in South Africa. | The night sky above Nairobi looks complete to the naked eye, but telescopes reveal it is only a sliver of what is actually there — limited by the small aperture of the human pupil and by our eyes' narrow sensitivity to visible light alone. Telescopes overcome both limits: larger apertures gather far more light from faint distant objects, and instruments tuned to infrared, ultraviolet, or radio wavelengths reveal sources our eyes cannot detect at all. Because light takes time to travel, every telescope is also a time machine — the JWST sees galaxies as they were just 300 million years after the Big Bang, evidence we could never reach by travelling. This is how humanity has built a detailed scientific account of a universe we cannot physically visit, with Africa contributing through MeerKAT and the SKA project in South Africa. |
| <b>Lesson 8</b><br><b>History of Space Exploration: From Sputnik to the James Webb Telescope — and Kenya's Place in the Cosmos</b> | Students examined 7 Mission Data Cards (Sputnik 1, Vostok 1, Apollo 11, Voyager 1 & 2, Hubble Space Telescope, Mars rovers, James Webb Space Telescope) and a briefing on the Kenya Space Agency (KSA) and the Square Kilometre Array (SKA) in South Africa. They recorded what each mission discovered and how that discovery contributes evidence toward the unit's driving question about the night sky above Nairobi.                                                                                                                                                                                                                                   | Students learned that human knowledge of the universe is not based on direct travel or touch, but on an incremental chain of evidence gathered by instruments — from Sputnik's proof of orbit in 1957, to the Hubble deep field images confirming cosmic expansion, to the JWST's infrared imaging of galaxies from only 300 million years after the Big Bang. They also learned that Kenya launched its first satellite (1KUNS-PF) from the ISS in 2018 and that Africa hosts the SKA — the world's largest radio telescope — positioning the continent as an active participant in answering the deepest questions about the universe.                                                                                                                                                                                                                        | Students explained, in their 3-sentence Evidence Summaries and updated cumulative models, that each space mission provided a specific type of evidence (light waves, radio waves, rock samples, infrared imaging) that revealed a specific aspect of the universe's history or structure. By adding a 'Human Evidence Layer' to their models — with arrows connecting missions to the parts of the universe each revealed — students demonstrated that our knowledge of what we see in the Nairobi night sky is the product of over 65 years of instrument-based, indirect evidence-gathering by scientists from many nations, including Kenya and Africa.                                                                                                                                                                                                           |
| <b>Lesson 9</b>                                                                                                                    | We observed the Sun emitting a coronal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | We learned that the Sun produces two                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | We explained that space is not a passive                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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| <b>Space Phenomena and Effects on Earth: Solar Flares, Geomagnetic Storms, Eclipses, and Tidal Forces</b> | mass ejection on May 10, 2024, in footage from NASA Solar Dynamics Observatory, and we observed documented evidence that this storm disrupted radio communications across East Africa, was recorded by the Mbita Point Research and Training Institute, and caused a measurable tidal anomaly in Mombasa — all traceable to a single event 150 million kilometres away on the Sun.                                                                                                                                               | types of emissions during a solar storm: electromagnetic radiation (arriving at Earth in 8 minutes at the speed of light) and a coronal mass ejection particle cloud (arriving 17 hours later). Earth is partially protected by its magnetosphere, but strong CMEs compress it and allow charged particles to disrupt radio communications, satellite signals, and power infrastructure. Eclipses result from precise geometric alignments of Sun, Moon, and Earth, and tidal forces from both the Moon and Sun affect coastal and lake communities in Kenya.                                                             | backdrop to life on Earth but an active system whose events — solar flares, geomagnetic storms, eclipses, and tidal forces — directly affect Kenyan infrastructure, communications, agriculture, and fishing. The universe visible in the night sky above Nairobi is not static or separate from us: it reaches down and shapes our daily lives, which is the most direct evidence yet that the driving question — how did everything in the night sky come to exist and how does it affect us — has a deeply personal answer for every Kenyan.                                                                                                                                    |
| <b>Lesson 10<br/>Careers in Space<br/>Physics: African and Kenyan Voices in the Cosmos</b>                | Students observed curated career profiles of real African and Kenyan space professionals across six career clusters (astrophysics, satellite engineering, meteorology, space policy, data science, telecommunications). They observed gender representation data showing approximately 28% of Kenya's space sector professionals are women, and they observed real institutional names — Kenya Space Agency, Kenya Meteorological Department, University of Nairobi, Strathmore University — connected to the careers discussed. | Students learned that the scientific questions anchoring this unit — how the universe began, how celestial bodies move, how we gather evidence about the cosmos — are actively investigated by real Kenyan and African professionals today. They learned specific career pathways, educational requirements, and institutional homes for space careers in Kenya. They learned that gender is not a scientific barrier to a space career, and that the Kenya Space Agency actively recruits women. They added a 'Human Network' layer to their cumulative universe model, placing each career at its correct cosmic scale. | Students explained, through Ted-Talk pitches and individual Exit Ticket Reflections, how each career cluster connects to specific evidence gathered in Lessons 1–9 of this unit. They explained, using the structured PCI discussion, why gender disparities in the space sector reflect social myth rather than scientific ability. They updated the Driving Question Board with new knowledge, new questions, and explicit cross-lesson connections — advancing collective progress towards answering: 'How did everything we see in the night sky above Nairobi come to exist, and how do scientists gather evidence about a universe they cannot directly touch or travel to?' |
| <b>Lesson 11<br/>Consolidation and Evidence Review:<br/>Building a Coherent Narrative of the Universe</b> | What evidence from across all 10 previous lessons did learners draw on today — including Big Bang evidence, cosmic expansion data, stellar formation, Solar System structure, planetary motion, and the methods scientists use to gather indirect evidence about the universe?                                                                                                                                                                                                                                                   | How does each of the 10 lessons contribute a distinct, necessary piece of evidence that together builds a complete explanation of how the universe — and everything visible in the night sky above Nairobi — came to exist and continues to evolve?                                                                                                                                                                                                                                                                                                                                                                       | How does the full body of evidence gathered across this unit answer the driving question: the night sky above Nairobi looks static and eternal, yet science shows it erupted from a hot dense point 13.8 billion years ago and has been expanding, evolving, and structuring itself ever since — and scientists know this not by travelling to the universe but by reading the light, radiation, and motion it sends to us here on Earth?                                                                                                                                                                                                                                          |
| <b>Lesson 12<br/>Model Building and Final Presentation: From the</b>                                      | Learners re-examined their Lesson 1 naïve models of the night sky above Nairobi and annotated them to identify conceptual                                                                                                                                                                                                                                                                                                                                                                                                        | Learners synthesised the full unit narrative: the universe began approximately 13.8 billion years ago in the Big Bang, producing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Learners constructed and defended a complete, evidence-based Final Explanation for the anchoring phenomenon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

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| <b>Big Bang to the Night Sky Above Nairobi</b> | <p>change. They audited their cumulative visual models against a 10-item Evidence Checklist covering the Big Bang, CMB radiation, stellar evolution, Solar System structure, gravitational forces, orbital mechanics, telescopic evidence, and African connections to space science. They participated in a structured CER Gallery Walk, delivering Final Explanation presentations and providing written peer feedback. They categorised all Driving Question Board questions using a traffic-light protocol. They wrote an individual Final Explanation and contributed to a whole-class synthesis explanation built sentence-by-sentence on the whiteboard.</p> | <p>space, time, matter, and energy; the first stars formed through gravitational collapse of hydrogen clouds; stellar evolution produced heavier elements distributed by supernovae; the Solar System formed from a collapsing nebula ~4.6 billion years ago; Newton's Law of Universal Gravitation governs orbital mechanics, affecting life in Kenya through tides, seasons, and the lunar cycle; scientists gather evidence about a universe they cannot travel to using CMB radiation, redshift observations, radio telescopes, and space probes; the night sky above Nairobi — seen from the Rift Valley or the shores of Lake Victoria — is not static but is a snapshot of a dynamic, evolving universe whose light carries billions of years of history.</p> | <p>and the unit Driving Question: the apparently permanent and unchanging night sky above Nairobi is in fact a window into 13.8 billion years of cosmic history. Every star is a sun undergoing nuclear fusion; the Milky Way is one of hundreds of billions of galaxies formed from primordial hydrogen; the planets orbit the Sun because of gravitational forces first mathematically described by Newton; the atoms in every Kenyan's body were forged in the cores of long-dead stars. Scientists know this not by travelling to the universe but by decoding the light, radiation, and gravitational signals it sends — using telescopes, satellites, and space probes that extend human senses across cosmic distances. The universe is not eternal and unchanging: it had a beginning, it has evolved, and it is still expanding today.</p> |
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## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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